Name

Mathematics Extension 1 Year 11

Compound Angle Identities

Book 1

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Syllabus Content

ME1-11-03 solves problems in three dimensions using trigonometry and simplifies expressions, proves results and solves problems involving compound angles using trigonometric identities

Further trigonometric identities

- Derive the sum and difference expansions for the trigonometric functions $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$, $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$ and $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$
- Use sum and difference expansion formulas to solve problems and prove results
- Derive the double angle formulas $\sin 2A = 2 \sin A \cos A$, $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$ and $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$
- Use the double angle formulas to solve problems and prove results

Compound Angle Identities

ⓘ Compound Angle Identities

$\sin(A+B) = \sin A \cos B + \cos A \sin B$

$\sin(A-B) = \sin A \cos B - \cos A \sin B$

$\cos(A+B) = \cos A \cos B - \sin A \sin B$

$\cos(A-B) = \cos A \cos B + \sin A \sin B$

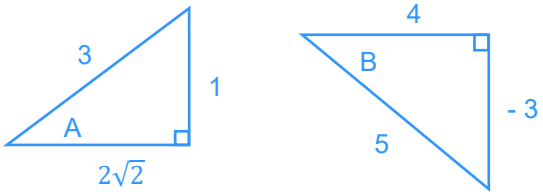
$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

Example Apply compound angle identities

Worked Example	Your Turn
Expand and simplify: $\sin(x - 30°)$ $= \sin x \cos 30° - \cos x \sin 30°$ $= \frac{\sqrt{3}\sin x}{2} - \frac{\cos x}{2}$	Expand and simplify: $\tan(A + 45°)$ $\frac{1+\tan A}{1-\tan A}$
Prove $\sin(90° + A) = \cos A$ $LHS = \sin 90° \cos A + \cos 90° \sin A$ $= 1(\cos A) + 0(\sin A)$ $= \cos A$ $= RHS$	Prove $2\cos\left(\theta + \frac{\pi}{3}\right) = \cos\theta - \sqrt{3}\sin\theta$
Find the exact value of $\cos 75°$. $\cos 75° = \cos(30° + 45°)$ $= \cos 30° \cos 45° - \sin 30° \sin 45°$ $= \frac{\sqrt{3}}{2}\left(\frac{1}{\sqrt{2}}\right) - \frac{1}{2}\left(\frac{1}{\sqrt{2}}\right)$ $= \frac{\sqrt{3}-1}{2\sqrt{2}}$	Find the exact value of $\sin 75°$. $\frac{\sqrt{2}+\sqrt{6}}{4}$

Example Apply Compound Angle Identities

Worked Example	Your Turn
A is acute and B is $-\frac{\pi}{2} < B < 0$. $\sin A = \frac{1}{3}$ and $\cos \beta = \frac{4}{5}$. Find the exact value of $\sin(A - B)$.	θ is acute and ϕ is $\pi < \phi < \frac{3\pi}{2}$. $\sin \theta = \frac{3}{5}$ and $\tan \phi = \frac{24}{7}$. Find the exact value of $\cos(\theta - \phi)$.
1. Draw triangles for A & B in relevant quadrant 	1. Triangles
2. Determine ratios and their sign. $\sin A = \frac{1}{3}$ $\sin B = -\frac{3}{5}$ $\cos A = \frac{2\sqrt{2}}{3}$ $\cos B = \frac{4}{5}$ $\tan A = \frac{1}{2\sqrt{2}}$ $\tan B = -\frac{3}{4}$	2. Ratios and sign
3. Find value using identity. $\sin(A - B) = \sin A \cos B - \cos A \sin B$ $= \frac{1}{3}\left(\frac{4}{5}\right) - \left(\frac{2\sqrt{2}}{3}\right)\left(-\frac{3}{5}\right)$ $= \frac{4}{15} + \frac{6\sqrt{2}}{15}$ $= \frac{4 + 6\sqrt{2}}{15}$	3. Find value $-\frac{4}{5}$

1 Expand using the compound angle identities.

a $\sin(x-y)$

sin x cos y − cos x sin y

b $\cos(2A+3B)$

cos 2A cos 3B − sin 2A sin 3B

c $\sin(3\alpha+5\beta)$

sin 3α cos 5β + cos 3α sin 5β

d $\cos\left(\theta-\frac{\phi}{2}\right)$

$\cos\theta\cos\frac{\phi}{2}+\sin\theta\sin\frac{\phi}{2}$

e $\tan(A+2B)$

$\frac{\tan A+\tan 2B}{1-\tan A\tan 2B}$

f $\tan(3\alpha-4\beta)$

$\frac{\tan 3\alpha-\tan 4\beta}{1+\tan 3\alpha\tan 4\beta}$

2 Express as a single trigonometric function.

a $\cos x\cos y-\sin x\sin y$

cos (x+y)

b $\sin 3\alpha\cos 2\beta+\cos 3\alpha\sin 2\beta$

sin (3α+2β)

c $\frac{\tan 40°-\tan 20°}{1+\tan 40°\tan 20°}$

tan 20°

d $\sin 5A\cos 2A-\cos 5A\sin 2A$

sin 3A

e $\cos 70°\cos 20°+\sin 70°\sin 20°$

cos 50°

f $\frac{\tan\alpha+\tan 10°}{1-\tan\alpha\tan 10°}$

tan (α+10°)

3 Use the compound-angle formulas to prove these rules:

a $\sin(90°+A)=\cos A$

= sin 90° cos A + cos 90° sin A
= 1·cos A + 0·sin A
= cos A

b $\cos(90°-A)=\sin A$

= cos 90° cos A + sin 90° sin A
= 0·cos A + 1·sin A
= sin A

c $\tan(360°-A)=-\tan A$

$=\frac{\tan 360°-\tan A}{1+\tan 360°\tan A}$
$=\frac{\tan 360°-\tan A}{1+\tan 360°\tan A}$
$=\frac{0-\tan A}{1+0}$
$=-\tan A$

d $\tan(180°+A)=\tan A$

$=\frac{\tan 180°+\tan A}{1-\tan 180°\tan A}$
$=\frac{0+\tan A}{1-0}$
$=\tan A$

e $\cos(270°-A)=-\sin A$

= cos 270° cos A + sin 270° sin A
= 0·cos A + (-1)·sin A
= -sin A

f $\sin(360°-A)=-\sin A$

$=\sin 360°\cos A-\cos 360°\sin A$
$=0\cdot\cos A-1\cdot\sin A$
$=-\sin A$

4 Prove these identities:

a $\sin(A+45°) = \frac{1}{\sqrt{2}}(\sin A + \cos A)$

LHS

$= \sin A \cos 45° + \cos A \sin 45°$

$= \sin A \left(\frac{1}{\sqrt{2}}\right) + \cos A \left(\frac{1}{\sqrt{2}}\right)$

$= \left(\frac{1}{\sqrt{2}}\right)(\sin A + \cos A) = \text{RHS}$

b $2\cos\left(\theta + \frac{\pi}{3}\right) = \cos\theta - \sqrt{3}\sin\theta$

LHS

$= 2\left(\cos\theta \cos\left(\frac{\pi}{3}\right) - \sin\theta \sin\left(\frac{\pi}{3}\right)\right)$

$= 2\left(\cos\theta \left(\frac{1}{2}\right) - \sin\theta\left(\frac{\sqrt{3}}{2}\right)\right)$

$= \cos\theta - \sqrt{3}\sin\theta = \text{RHS}$

c $\tan\left(\frac{\pi}{4} - A\right) = \frac{1-\tan A}{1+\tan A}$

LHS

$= \frac{\tan\left(\frac{\pi}{4}\right) - \tan A}{1 + \tan\left(\frac{\pi}{4}\right)\tan A}$

$= \frac{1 - \tan A}{1 + \tan A} = \text{RHS}$

d $\sin(A-30°) = \frac{1}{2}(\sqrt{3}\sin A - \cos A)$

LHS

$= \sin A \cos 30° - \cos A \sin 30°$

$= \sin A \left(\frac{\sqrt{3}}{2}\right) - \cos A \left(\frac{1}{2}\right)$

$= \left(\frac{1}{2}\right)(\sqrt{3}\sin A - \cos A) = \text{RHS}$

5

a If $\tan\alpha = \frac{1}{2}$, $\tan\beta = \frac{1}{3}$, find $\tan(\alpha+\beta)$

$\tan(\alpha + \beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$

$= \frac{\frac{1}{2}+\frac{1}{3}}{1 - \left(\frac{1}{2}\right)\left(\frac{1}{3}\right)}$

$= \frac{\frac{5}{6}}{1 - \frac{1}{6}}$

$= \frac{\frac{5}{6}}{\frac{5}{6}}$

$= 1$

b If $\cos A = \frac{4}{5}$, $\sin B = \frac{12}{13}$, where A, B are acute, find $\sin(A+B)$

$\sin(A + B) = \sin A \cos B + \cos A \sin B$

$= \left(\frac{3}{5}\right)\left(\frac{5}{13}\right) + \left(\frac{4}{5}\right)\left(\frac{12}{13}\right)$

$= \frac{15}{65} + \frac{48}{65}$

$= \frac{63}{65}$

c If $\sin\theta = \frac{2}{3}$, $\cos\phi = \frac{1}{4}$, where θ, ϕ are acute, find $\cos(\theta - \phi)$

$\cos(\theta - \varphi) = \cos\theta\cos\varphi + \sin\theta\sin\varphi$

$= (\sqrt{5}/3)(1/4) + \left(\frac{2}{3}\right)\left(\frac{\sqrt{15}}{4}\right)$

$= \frac{\sqrt{5}}{12} + \frac{2\sqrt{15}}{12}$

$= \frac{\sqrt{5} + 2\sqrt{15}}{12}$

$= \frac{\sqrt{5}(1+2\sqrt{3})}{12}$

6 Prove these identities

a $\sin(A+B) + \sin(A-B) = 2\sin A\cos B$

LHS = (sin A cos B + cos A sin B) + (sin A cos B - cos A sin B)

= 2 sin A cos B = RHS

b $\cos(x-y) - \cos(x+y) = 2\sin x \sin y$

LHS = (cos x cos y + sin x sin y) - (cos x cos y - sin x sin y)

= 2 sin x sin y = RHS

c $\sin(x+y) + \cos(x-y) = (\sin x + \cos x)(\sin y + \cos y)$

LHS = sin x cos y + cos x sin y + cos x cos y + sin x sin y

= sin x(cos y + sin y) + cos x(sin y + cos y)

= (sin x + cos x)(sin y + cos y) = RHS

7 By expressing $15°$ as $(45° - 30°)$, show:

a $\sin 15° = \dfrac{\sqrt{3}-1}{2\sqrt{2}}$

$$\begin{aligned}\sin 15° &= \sin(45° - 30°)\\ &= \sin 45° \cos 30° - \cos 45° \sin 30°\\ &= \left(\frac{1}{\sqrt{2}}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{2}\right)\\ &= \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}}\\ &= \frac{\sqrt{3}-1}{2\sqrt{2}}\end{aligned}$$

b $\cos 15° = \dfrac{\sqrt{3}+1}{2\sqrt{2}}$

$$\begin{aligned}\cos 15° &= \cos(45° - 30°)\\ &= \cos 45° \cos 30° + \sin 45° \sin 30°\\ &= (1/\sqrt{2})(\sqrt{3}/2) + \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{2}\right)\\ &= \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}}\\ &= \frac{\sqrt{3}+1}{2\sqrt{2}}\end{aligned}$$

c $\tan 15° = 2 - \sqrt{3}$

$$\begin{aligned}\tan 15° &= \frac{\sin 15°}{\cos 15°}\\ &= \frac{\left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)}{\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)}\\ &= \frac{\sqrt{3}-1}{\sqrt{3}+1}\\ &= \frac{3-2\sqrt{3}+1}{3-1}\\ &= \frac{4-2\sqrt{3}}{2}\\ &= 2-\sqrt{3}\end{aligned}$$

8 Find the following in exact form:

a $\sin 75°$

$$\begin{aligned}\sin 75° &= \sin(45° + 30°)\\ &= \sin 45° \cos 30° + \cos 45° \sin 30°\\ &= (1/\sqrt{2})(\sqrt{3}/2) + \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{2}\right)\\ &= \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}}\\ &= \frac{\sqrt{3}+1}{2\sqrt{2}}\end{aligned}$$

b $\cos 75°$

$$\begin{aligned}\cos 75° &= \cos(45° + 30°)\\ &= \cos 45° \cos 30° - \sin 45° \sin 30°\\ &= (1/\sqrt{2})(\sqrt{3}/2) - \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{2}\right)\\ &= \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}}\\ &= \frac{\sqrt{3}-1}{2\sqrt{2}}\end{aligned}$$

c $\tan 75°$

$$\begin{aligned}\tan 75° &= \sin 75°/\cos 75°\\ &= \frac{\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)}{\left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)}\\ &= \frac{\sqrt{3}+1}{\sqrt{3}-1}\\ &= \frac{3+2\sqrt{3}+1}{3-1}\\ &= \frac{4+2\sqrt{3}}{2}\\ &= 2+\sqrt{3}\end{aligned}$$

9 If $\sin A = \frac{2}{3}$, where $\frac{\pi}{2} < A < \pi$, and $\tan B = \frac{2}{3}$, where $\pi < B < \frac{3\pi}{2}$, show that:

$$\cos(A+B) = \frac{3\sqrt{5}+4}{3\sqrt{13}}$$

By drawing a triangle in Q2:

$$\cos A = -\frac{\sqrt{5}}{3}$$

By drawing a triangle in Q3:

$$\sin B = -\frac{2}{\sqrt{13}}, \cos B = -\frac{3}{\sqrt{13}}$$

cos(A + B) = cos A cos B – sin A sin B

$$\begin{aligned}&= \left(-\frac{\sqrt{5}}{3}\right)\left(-\frac{3}{\sqrt{13}}\right) - \left(\frac{2}{3}\right)\left(-\frac{2}{\sqrt{13}}\right)\\ &= \frac{3\sqrt{5}}{3\sqrt{13}} + \frac{4}{3\sqrt{13}} = \frac{3\sqrt{5}+4}{3\sqrt{13}}\end{aligned}$$

10 Use compound angle formulae to find the exact value of:

a $\cos 105°$

use $105° = 60° + 45°$

$= \frac{1 - \sqrt{3}}{2\sqrt{2}}$

b $\sin \frac{13\pi}{12}$

Use $\frac{13\pi}{12} = \frac{5\pi}{6} + \frac{\pi}{4}$

This is easier if you convert to degrees!

$\frac{\sqrt{3} - 1}{2\sqrt{2}}$

c $\cot 285°$

Use 285° = 240° + 45°

$\sqrt{3} - 2$

11

a Show that $\sin(A + B)\sin(A - B) = \sin^2 A - \sin^2 B$

LHS

= sin(A + B) sin(A − B)

= (sin A cos B + cos A sin B)(sin A cos B − cos A sin B)

= sin² A cos² B − cos² A sin² B

= sin² A (1 − sin² B) − (1 − sin² A) sin² B

= sin² A − sin² A sin² B − sin² B + sin² A sin² B

= sin² A − sin² B = RHS

b Hence simplify $\sin^2 75° - \sin^2 15°$

$\sin^2 75° - \sin^2 15°$

$= \sin(75° + 15°)\sin(75° - 15°)$

$= \sin 90° \sin 60°$

$= \frac{\sqrt{3}}{2}$

12 Prove these identities

a $\frac{2\sin(x-y)}{\cos(x+y) - \cos(x-y)} = \cot x - \cot y$

LHS

$= \frac{2\sin x \cos y - 2\cos x \sin y}{\cos x \cos y - \sin x \sin y - (\cos x \cos y + \sin x \sin y)}$

$= \frac{2\sin x \cos y - 2\cos x \sin y}{-2\sin x \sin y}$

$= \frac{2\sin x \cos y}{-2\sin x \sin y} - \frac{2\cos x \sin y}{-2\sin x \sin y}$

$= -\frac{\cos y}{\sin y} + \frac{\cos x}{\sin x}$

$= -\cot y + \cot x$

= RHS

b $\frac{\sin(\theta+\phi)}{\sin(\theta-\phi)} = \frac{\tan\theta + \tan\phi}{1 + \tan\theta \tan\phi}$

LHS

$= \frac{\sin\theta\cos\phi + \cos\theta\sin\phi}{\cos\theta\cos\phi + \sin\theta\sin\phi}$

Divide numerator and denominator by $\cos\theta\cos\phi$

$= \frac{\tan\theta + \tan\phi}{1 + \tan\theta\tan\phi}$

13 Prove $\frac{\sin(\alpha-\beta)}{\sin\alpha\sin\beta}+\frac{\sin(\beta-\gamma)}{\sin\beta\sin\gamma}+\frac{\sin(\gamma-\alpha)}{\sin\gamma\sin\alpha}=0$

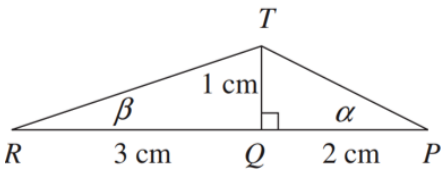
$$\text{LHS}=\frac{\sin\alpha\cos\beta-\cos\alpha\sin\beta}{\sin\alpha\sin\beta}+\frac{\sin\beta\cos\gamma-\cos\beta\sin\gamma}{\sin\beta\sin\gamma}+\frac{\sin\gamma\cos\alpha-\cos\gamma\sin\alpha}{\sin\gamma\sin\alpha}$$

Split each fraction and simplify

$$=(\cot\beta-\cot\alpha)+(\cot\gamma-\cot\beta)+(\cot\alpha-\cot\gamma)$$

$$=0$$

14 Consider the triangle.



a Write the formula for $\sin(\alpha+\beta)$.

$$\sin\alpha\cos\beta+\cos\alpha\sin\beta$$

b Hence, using the diagram, show that the angles α and β sum to 45°.

$$\sin(\alpha+\beta)=\sin\alpha\cos\beta+\cos\alpha\sin\beta$$

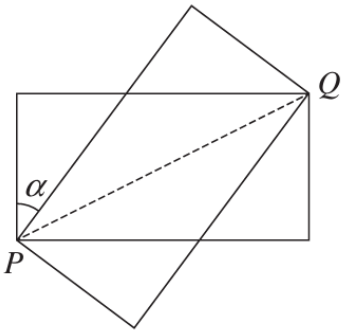
$$=\frac{1}{\sqrt{5}}\left(\frac{3}{\sqrt{10}}\right)+\frac{2}{\sqrt{5}}\left(\frac{1}{\sqrt{10}}\right)$$

$$=\frac{1}{\sqrt{2}}$$

$$\therefore\ \alpha+\beta=\sin^{-1}\frac{1}{\sqrt{2}}=45°$$

15 Consider the diagram. Each rectangle is 6 cm by 3 cm and share a common diagonal PQ.

Show that $\tan\alpha=\frac{3}{4}$



$$PQ=\sqrt{6^2+3^2}=3\sqrt{5}$$

Let angle complementary of α be β

$$\tan\beta=\frac{3}{6}=\frac{1}{2}$$

$$\tan(\alpha+\beta)=\frac{6}{3}=2$$

$$\tan(\alpha+\beta)=\frac{\tan\alpha+\tan\beta}{1-\tan\alpha\tan\beta}$$

$$2=\frac{\tan\alpha+\frac{1}{2}}{1-\frac{1}{2}\tan\alpha}$$

$$2-\tan\alpha=\tan\alpha+\frac{1}{2}$$

$$2\tan\alpha=\frac{3}{2}$$

$$\tan\alpha=\frac{3}{4}$$

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a The numbers A, B, and C are related by the equations $A = B - d$ and $C = B + d$, where d is a real number.

Show that $\frac{\sin A+\sin C}{\cos A+\cos C} = \tan B$

$$\frac{\sin A + \sin C}{\cos A + \cos C} = \frac{\sin(B-d) + \sin(B+d)}{\cos(B-d) + \cos(B+d)}$$
$$= \frac{2\sin B\cos d}{2\cos B\cos d}$$
$$= \frac{\sin B}{\cos B} = \tan B$$

b Hence, or otherwise, solve $\frac{\sin\frac{5\theta}{7}+\sin\frac{6\theta}{7}}{\cos\frac{5\theta}{7}+\cos\frac{6\theta}{7}} = \sqrt{3}$ for $0 \le \theta \le 2\pi$.

$$Let\ A = \frac{5\theta}{7}, C = \frac{6\theta}{7}$$
$$\text{using } d = B - A \text{ and } d = C - B$$
$$B - A = C - B$$
$$\therefore B = \frac{A + C}{2}$$
$$= \frac{\frac{5\theta}{7} + \frac{6\theta}{7}}{2}$$
$$= \frac{11\theta}{14}$$

$$\text{Using } \frac{\sin A+\sin C}{\cos A+\cos C} = \tan B$$
$$\frac{\sin\frac{5\theta}{7}+\sin\frac{6\theta}{7}}{\cos\frac{5\theta}{7}+\cos\frac{6\theta}{7}} = \tan\frac{11\theta}{14} = \sqrt{3}$$
$$\frac{11\theta}{14} = \frac{\pi}{3}, \frac{4\pi}{3}$$
$$\theta = \frac{14\pi}{33}, \frac{56\pi}{33}$$

Double Angle Identities

ⓘ Double Angle Identities

$\sin(2\theta) = 2\sin\theta\cos\theta$

$\cos(2\theta) = \cos^2\theta - \sin^2\theta$

$\tan(2\theta) = \dfrac{2\tan\theta}{1-\tan^2\theta}$

$\cos(2\theta) = \cos^2\theta - \sin^2\theta$

$= 2\cos^2\theta - 1$

$= 1 - 2\sin^2\theta$

Example Apply double angle identities

Worked Example

Prove

$(\sin x + \cos x)^2 = 1 + \sin 2x$

$LHS = (\sin x + \cos x)^2$

$= \sin^2 x + 2\sin x\cos x + \cos^2 x$

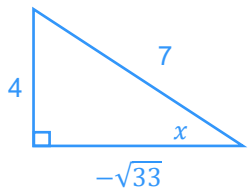
$= 1 + 2\sin x$

$= RHS$

If $\sin x = \frac{4}{7}$, where x is obtuse,

Find $\sin 2x$.

1. Draw triangle in correct quadrant.



2. Determine ratios and their sign.

$\sin x = \frac{4}{7}\cos x = -\frac{\sqrt{33}}{7}$

$\tan x = -\frac{4}{\sqrt{33}}$

3. Find value using identity.

$\sin 2x = 2\sin x\cos x$

$\sin 2x = 2\left(\frac{4}{7}\right)\left(-\frac{\sqrt{33}}{7}\right)$

$= -\frac{8\sqrt{33}}{49}$

Your Turn

Prove

$\cot 2x = \dfrac{\cot^2 x - 1}{2\cot x}$

(hint: divide by $\tan^2 x$)

If $\sin\alpha = -\frac{3}{5}$, where $-\frac{\pi}{2} < \alpha < 0$,

Find $\tan 2\alpha$.

$-\frac{12}{7}$

1 Use the double-angle identities to simplify:

a $2\sin x\cos x$

sin 2x

b $\cos^2\theta - \sin^2\theta$

cos 2θ

c $\dfrac{2\tan\alpha}{1-\tan^2\alpha}$

tan 2α

d $2\sin 20°\cos 20°$

sin 40°

e $2\cos^2 50° - 1$

cos 100°

f $\dfrac{2\tan 70°}{1-\tan^2 70°}$

tan 140°

g $2\sin 3\theta\cos 3\theta$

sin 6θ

h $1 - 2\sin^2 2A$

cos 4A

i $\dfrac{2\tan 4x}{1-\tan^2 4x}$

tan 8x

2 Prove that:

a $(\cos A - \sin A)(\cos A + \sin A) = \cos 2A$

LHS
= cos² A - sin² A
= cos 2A
= RHS

b $(\sin\alpha - \cos\alpha)^2 = 1 - \sin 2\alpha$

LHS
= sin² α - 2 sin α cos α + cos² α
= (sin² α + cos² α) - 2 sin α cos α
= 1 - 2 sin α cos α
= 1 - sin 2α = RHS

c $\sin 2\theta = 2\sin\theta\sin\left(\frac{\pi}{2} - \theta\right)$

RHS
$= 2\sin\theta\sin\left(\frac{\pi}{2} - \theta\right)$
$= 2\sin\theta\cos\theta$
$= \sin 2\theta$
$=$ LHS

d $\dfrac{1}{1-\tan\theta} - \dfrac{1}{1+\tan\theta} = \tan 2\theta$

LHS
$= \dfrac{1 + \tan\theta - (1 - \tan\theta)}{(1 - \tan\theta)(1 + \tan\theta)}$
$= \dfrac{2\tan\theta}{1 - \tan^2\theta}$
$= \tan 2\theta$
$=$ RHS

3

a If $\cos\alpha = \frac{4}{5}$, find $\cos 2\alpha$

$\cos 2\alpha = 2\cos^2\alpha - 1$
$= 2\left(\frac{4}{5}\right)^2 - 1$
$= \frac{7}{25}$

b If $\sin x = \frac{2}{3}$, find $\cos 2x$

$\cos 2x = 1 - 2\sin^2 x$
$= 1 - 2\left(\frac{2}{3}\right)^2$
$= \frac{1}{9}$

c If $\sin\theta = \frac{5}{13}$ and θ is acute, find $\sin 2\theta$

$\sin 2\theta = 2\sin\theta\cos\theta$
$= 2\left(\frac{5}{13}\right)\left(\frac{12}{13}\right)$
$= \frac{120}{169}$

d If $\tan A = \frac{1}{2}$, find $\tan 2A$

$\tan 2A = 2\tan A/(1 - \tan^2 A)$
$= \dfrac{2\left(\frac{1}{2}\right)}{1 - \left(\frac{1}{2}\right)^2}$
$= \frac{4}{3}$

4 If $\sin x = \frac{3}{4}$ and $\frac{\pi}{2} < x < \pi$, find the exact value of $\sin 2x$.

By drawing a right–angled triangle in Q2:

$\cos x = -\frac{\sqrt{7}}{4}$

$\sin 2x = 2\sin x \cos x$

$= 2\left(\frac{3}{4}\right)\left(-\frac{\sqrt{7}}{4}\right)$

$= -\frac{3\sqrt{7}}{8}$

5 Prove each identity.

a $\cos^4 \alpha - \sin^4 \alpha = \cos 2\alpha$

LHS

$= (\cos^2 \alpha - \sin^2 \alpha)(\cos^2 \alpha + \sin^2 \alpha)$

$= \cos^2 \alpha - \sin^2 \alpha$

$= \cos 2\alpha$

$= RHS$

b $\cos 2x + \cos x = (\cos x + 1)(2\cos x - 1)$

LHS

$= 2\cos^2 x - 1 + \cos x$

$= 2\cos^2 x + \cos x - 1$

$= (2\cos x - 1)(\cos x + 1)$

$= RHS$

c $\frac{\sin 2A}{1-\cos 2A} = \cot A$

LHS

$= \frac{2\sin A \cos A}{1-(1-2\sin^2 A)}$

$= \frac{\sin A \cos A}{\sin^2 A}$

$= \frac{\cos A}{\sin A}$

$= \cot A$

$= RHS$

d $\cos\theta - \sin\theta \sin 2\theta = \cos\theta \cos 2\theta$

LHS

$= \cos\theta - 2\sin^2\theta \cos\theta$

$= \cos\theta\,(1 - 2\sin^2\theta)$

$= \cos\theta \cos 2\theta$

$= RHS$

e $\tan\left(\frac{\pi}{4} + \alpha\right) - \tan\left(\frac{\pi}{4} - \alpha\right) = 2\tan 2\alpha$

LHS

$= \frac{\tan\frac{\pi}{4} + \tan\alpha}{1 - \tan\frac{\pi}{4}\tan\alpha} - \frac{\tan\frac{\pi}{4} - \tan\alpha}{1 + \tan\frac{\pi}{4}\tan\alpha}$

$= \frac{1+\tan\alpha}{1-\tan\alpha} - \frac{1-\tan\alpha}{1+\tan\alpha}$

$= \frac{(1+\tan\alpha)^2 - (1-\tan\alpha)^2}{(1-\tan\alpha)(1+\tan\alpha)}$

$= \frac{4\tan\alpha}{1-\tan^2\alpha}$

$= 2\tan 2\alpha$

$= RHS$

f $\frac{\sin 2\theta + \sin\theta}{\cos 2\theta + \cos\theta + 1} = \tan\theta$

LHS

$= \frac{2\sin\theta\cos\theta + \sin\theta}{\cos 2\theta + \cos\theta + 1}$

$= \frac{\sin\theta\,(2\cos\theta + 1)}{2\cos^2\theta + \cos\theta}$

$= \frac{\sin\theta\,(2\cos\theta + 1)}{\cos\theta\,(2\cos\theta + 1)}$

$= \tan\theta$

$= RHS$

6

a Show that $\frac{1-\cos 2x}{1+\cos 2x} = \tan^2 x$.

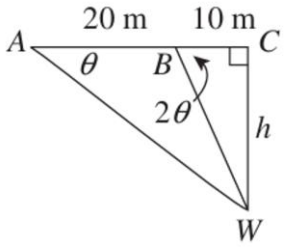
$$LHS = \frac{1-\cos 2x}{1+\cos 2x}$$
$$= \frac{1-(1-2\sin^2 x)}{1+(2\cos^2 x-1)}$$
$$= \frac{\sin^2 x}{\cos^2 x}$$
$$= \tan^2 x$$

b Hence find the exact value of $\tan\frac{\pi}{8}$.

Let $x = \frac{\pi}{8}$

$$\tan^2\frac{\pi}{8} = \frac{1-\cos\frac{\pi}{4}}{1+\cos\frac{\pi}{4}}$$
$$= \frac{1-\frac{1}{\sqrt{2}}}{1+\frac{1}{\sqrt{2}}}$$
$$= \frac{\sqrt{2}-1}{\sqrt{2}+1}$$
$$= (\sqrt{2}-1)^2$$
$$\therefore \tan\frac{\pi}{8} = \sqrt{2}-1$$

7 Consider the diagram.



a Write $\tan 2\theta$ in terms of $\tan\theta$.

$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

b Use the two right-angled triangles to write two equations with h and θ.

$$\tan\theta = \frac{h}{30}, \tan 2\theta = \frac{h}{10}$$

c Use parts **a** and **b** to show $\frac{1}{10} = \frac{60}{900-h^2}$

$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$
$$\frac{h}{10} = \frac{\frac{2h}{30}}{1-\left(\frac{h}{30}\right)^2}$$
$$\frac{1}{10} = \frac{\left(\frac{1}{15}\right)}{\left(1-\frac{h^2}{900}\right)}$$
$$\frac{1}{10} = \frac{\left(\frac{1}{15}\right)}{\left(\frac{900-h^2}{900}\right)}$$
$$\frac{1}{10} = \frac{1}{15}\times\frac{900}{900-h^2}$$
$$\frac{1}{10} = \frac{60}{900-h^2}$$

d Show that $h = 10\sqrt{3}$ metres.

$$900-h^2 = 600$$
$$h^2 = 300$$
$$h = \sqrt{300} = 10\sqrt{3}$$

8 Use double-angle formulas to show that:

a $2\sin\frac{4\pi}{5}\cos\frac{\pi}{5} = \sin\frac{2\pi}{5}$

LHS

$= 2\sin\frac{\pi}{5}\cos\frac{\pi}{5}$

$= \sin\frac{2\pi}{5}$

b $\cos^2\frac{4\pi}{7} - \sin^2\frac{3\pi}{7} = \cos\frac{6\pi}{7}$

LHS

$= \left(-\cos\frac{3\pi}{7}\right)^2 - \sin^2\frac{3\pi}{7}$

$= \cos^2\frac{3\pi}{7} - \sin^2\frac{3\pi}{7}$

$= \cos\frac{6\pi}{7}$

9

a By using $3\theta = 2\theta + \theta$ and using compound and double-angle results, prove that

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta$$

LHS

$= \cos 2\theta\cos\theta - \sin 2\theta\sin\theta$

$= (2\cos^2\theta - 1)\cos\theta - 2\sin^2\theta\cos\theta$

$= \cos^3\theta - \cos\theta - 2\cos\theta(1 - \cos^2\theta)$

$= \cos^3\theta - 3\cos\theta$

$= RHS$

b Hence solve $8\cos^3\theta - 6\cos\theta + 1 = 0$ in the domain $0 \le \theta \le 2\pi$.

Let $x = \cos\frac{2\pi}{3}$

$8x^3 - 6x + 1 = 0$

$4x^3 - 3x + \frac{1}{2} = 0$

Let $x = \cos\theta$

$4\cos^3\theta - 3\cos\theta + \frac{1}{2} = 0$

$\cos 3\theta + \frac{1}{2} = 0$

$\cos 3\theta = -\frac{1}{2}$

Solve trig equation in the domain $0 \le 3\theta \le 6\pi$

$3\theta = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \frac{10\pi}{3}, \frac{14\pi}{3}, \frac{16\pi}{3}$

$\theta = \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9}, \frac{10\pi}{9}, \frac{14\pi}{9}, \frac{16\pi}{9}$

10 Prove these identities.

a $\cot 2\alpha + \tan \alpha = \operatorname{cosec} 2\alpha$

LHS

$= \frac{\cos 2\alpha}{\sin 2\alpha} + \frac{\sin \alpha}{\cos \alpha}$

$= \frac{1 - 2\sin^2 \alpha + \sin^2 \alpha}{2 \sin \alpha \cos \alpha}$

$= \frac{1}{\sin 2\alpha}$

$= \operatorname{cosec} 2\alpha$

b $\frac{\sin 3A}{\sin A} + \frac{\cos 3A}{\cos A} = 4 \cos 2A$

LHS

$= \frac{\sin 3A \cos A + \cos 3A \sin A}{\sin A \cos A}$

$= \frac{\sin 4A}{\frac{1}{2}\sin 2A}$

$= \frac{2 \sin 2A \cos 2A}{\frac{1}{2}\sin 2A}$

$= 4 \cos 2A$

c $\tan 2x \cot x = 1 + \sec 2x$

LHS

$= \frac{\sin 2x \cos x}{\cos 2x \sin x}$

$= \frac{2 \sin x \cos^2 x}{\cos 2x \sin x}$

$= \frac{2 \sin x \left(\frac{1}{2}(1 + \cos 2x)\right)}{\cos 2x \sin x}$

$= \frac{\sin x + \sin x \cos 2x}{\cos 2x \sin x}$

$= \frac{1}{\cos 2x} + 1$

$= \sec 2x + 1$

d $\frac{\sin 2\theta - \cos 2\theta + 1}{\sin 2\theta + \cos 2\theta - 1} = \tan\left(\theta + \frac{\pi}{4}\right)$

LHS

$\frac{2 \sin \theta \cos \theta - (1 - 2\sin^2 \theta) + 1}{2 \sin \theta \cos \theta + (1 - 2\sin^2 \theta) - 1}$

$= \frac{2 \sin \theta \cos \theta + 2\sin^2 \theta}{2 \sin \theta \cos \theta - 2\sin^2 \theta}$

$= \frac{2 \sin \theta (\cos \theta + \sin \theta)}{2 \sin \theta (\cos \theta - \sin \theta)}$

$= \frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta}$

Divide numerator and denominator by cos θ

$= \frac{1 + \tan \theta}{1 - \tan \theta}$

RHS

$\frac{\tan \theta + \tan \frac{\pi}{4}}{1 - \tan \theta \tan \frac{\pi}{4}}$

$= \frac{\tan \theta + 1}{1 - \tan \theta}$

∴ LHS = RHS

e $\frac{1 + \sin 2\alpha}{1 + \cos 2\alpha} = \frac{1}{2}(1 + \tan \alpha)^2$

LHS

$= \frac{1 + 2 \sin \alpha \cos \alpha}{2 \cos^2 \alpha}$

$= \frac{1}{2 \cos^2 \alpha} + \frac{\sin \alpha}{\cos \alpha}$

$\frac{1}{2}\sec^2 \alpha + \tan \alpha$

$= \frac{1}{2}(1 + \tan^2 \alpha) + \tan \alpha$

$= \frac{1}{2} + \frac{1}{2}\tan^2 \alpha + \tan \alpha$

RHS

$= \frac{1}{2}(1 + 2 \tan \alpha + \tan^2 \alpha)$

$= \frac{1}{2} + \tan \alpha + \frac{1}{2}\tan^2 \alpha$

∴ LHS = RHS

f $\operatorname{cosec} 4A + \cot 4A = \frac{1}{2}(\cot A - \tan A)$

LHS

$\frac{1}{\sin 4A} + \frac{\cos 4A}{\sin 4A}$

$= \frac{1 + 2\cos^2 2A - 1}{2 \sin 2A \cos 2A}$

$= \frac{\cos 2A}{\sin 2A}$

$= \frac{\cos^2 A - \sin^2 A}{2 \sin A \cos A}$

RHS

$\frac{1}{2}\left(\frac{\cos A}{\sin A} - \frac{\sin A}{\cos A}\right)$

$= \frac{1}{2}\left(\frac{\cos^2 A - \sin^2 A}{\sin A \cos A}\right)$

$= \frac{\cos^2 A - \sin^2 A}{2 \sin A \cos A}$

∴ LHS = RHS